

Cyber Security Task Force: Progress Metrics

- **Current Progress:** 5% (Commencing Week 1, Day 2 of the 3-Month Bootcamp).
- **Current Focus:** Week 1 Mandate — Application Layer Vulnerabilities, HTTP Manipulation, and Input & Data Safety.

The Daily Drop Blueprint

Title: Cyber Security Task Force: Week 1, Tuesday - Input & Data Safety and Manual SQL Injection

Section A: Today's Materials (The Synthesis Mandate)

You must synthesize the information from all sources below to complete today's execution mandate. Do not consult outside tutorials.

1. The Video Drills (Watch These First)

- **Scrimba Anchor 1:** [Learn Cybersecurity - Module: "Input & Data Safety"](#) (Crucial for understanding how unsanitized data interacts with backend systems)[cite: 22].
- **External Video 1:** [YouTube: Computerphile - SQL Injection](#) (The definitive visual breakdown of how SQL databases process malicious strings).
- **External Video 2:** [YouTube: PortSwigger - SQL Injection Authentication Bypass](#) (Mandatory viewing for your manual exploitation phase).

2. Required Technical Documentation

- **Primary Architecture Doc:** [OWASP SQL Injection Prevention Cheat Sheet](#) (The enterprise standard for remediating your findings).
- **Specific Syntax/Rule Doc:** [PortSwigger Web Security: SQL Injection Cheat Sheet](#) (Required for crafting your payloads).
- **Configuration Doc:** [Burp Suite Official Docs: Intercepting HTTP Traffic](#) (Essential for manipulating hidden web parameters)[cite: 21].

3. Engineering Assets

- **Vulnerable Asset:** OWASP Juice Shop (Sandbox Application) (You must continue using your locally deployed Docker or Node.js instance from Day 1)[cite: 21, 23].
- **LLM Prompting Asset:** [OpenAI Official Guide: Prompt Engineering](#) (Use this to ask the LLM to explain the mechanics of a specific CVE or to summarize remediation logic)[cite: 21].
- **Reporting Asset:** Vulnerability Report Template (Google Docs) (Utilize the exact structure provided yesterday for your forensic documentation)[cite: 23].

Section B: The Execution Mandate (9:00 AM - 2:00 PM)

1. Execution Summary (TL;DR)

Objective: Route your browser traffic through Burp Suite to intercept and manipulate HTTP requests. You will manually craft SQL Injection payloads to bypass the OWASP Juice Shop authentication layer and document the exact database mechanics and remediation code in a forensic vulnerability report[cite: 21].

2. The Workflow:

- **9:00 AM - 10:00 AM (Learning):** Watch the SQL Injection and Input Safety drills. Read the OWASP documentation to understand why input sanitization alone is insufficient compared to parameterized queries.
- **10:00 AM - 1:00 PM (Building & Exploiting):**
 - **Action 1 (Proxy Setup):** Configure your web browser to route traffic through Burp Suite Community Edition[cite: 21]. Ensure you can intercept and modify outbound HTTP POST requests.
 - **Action 2 (Manual Exploitation):** Attempt to bypass the login screen of the OWASP Juice Shop. You must manually craft ' OR 1=1 -- payloads to trick the backend database into authenticating you as the administrator[cite: 21].
 - **Action 3 (Forensic Reporting):** Draft a formal Vulnerability Report. You must include the CVSS severity score, the exact Burp Suite HTTP request/response snippet (Proof of Concept), and the architectural fix required[cite: 21].
- **The Non-Negotiable Rules:**
 1. **The Automated Tool Ban:** You are strictly forbidden from using automated tools like SQLMap. You must craft the payloads manually to prove you understand what the -- (comment) syntax is doing to the backend SQL query[cite: 21].
 2. **The Remediation Logic Rule:** Finding the hole is only half the job. Your report cannot just state "Fix the SQLi." You must provide the exact secure coding pattern (e.g., parameterized queries or prepared statements) required to patch the vulnerability[cite: 21].
- **1:00 PM - 1:30 PM (Hygiene):** Verify your PoC reproduction steps are crystal clear. Push your markdown-formatted Vulnerability Report to your internal Security repository on GitHub.
- **1:30 PM - 2:00 PM (LinkedIn):** Draft a technical summary detailing the mechanics of SQL Injection. Explain the architectural difference between simple input sanitization and utilizing Prepared Statements. You must hyperlink your ProSensia admit card within the post text.



Section C: Submission Protocol (Due strictly by 2:00 PM)

- **Step 1 (The Kanban Board):** Verify that this specific task card is shifted to the "Under Review" column on your team board.
- **Step 2 (The Assignments Module):** Navigate directly to today's entry in the platform's independent Assignment block. Paste both the link to your security repository report and your live LinkedIn Post URL into the designated submission form fields.
- **Operational Note:** All grading and technical feedback loops operate through this portal. Do not DM Management your links[cite: 21, 23].